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Physical Characteristics of Le Sueur Clay Loam Soil Following No-till and Conventional Tillage[1]

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ABSTRACT

Although the effects of no-till on crop yields has been widely investigated, there are virtually no data on its physical effects on clay loam soils like those in the Corn Belt. In order to compare the effects of no-till with conventional fall plow tillage on a south-central Minnesota clay loam soil (Aquic Argiudoll), we measured some soil physical properties 6 years after beginning a field experiment with corn (*Zea mays* L.). Data show that soil under no-till had significantly greater bulk density, both in spring and fall, for samples in the surface 30 cm. Densities of no-till soil ranged from 1.24 to 1.32 g/cm^3 in in contrast to those of conventional tillage, which ranged from 1.05 to 1.12 g/cm^3. Air-filled porosities of surface samples under no-till were lower at all potentials measured. At —100 mb, no-till averaged 0.143 cm^3/cm^3 compared to 0.198 cm^3/cm^3 for conventional tillage. Mean number of channels 1 mm and greater created by earthworms and decomposed rootlets were significantly greater for no-till, ranging from 666 to 1,732/m^2 compared to 243 to 1,475/m^2 for conventional tillage. Volumetric water contents were also greater for no-till surface samples, ranging from 0.35 to 0.28 cm^3/cm^3 vs. 0.31 to 0.25 cm^3/cm^3 for conventional tillage. Saturated hydraulic conductivities of spring surface samples were lower under no-till than conventional tillage averaging 14.6 cm/hour and 38.2 cm/hour, respectively. Generally. differences in measured parameters between treatments were smaller for samples collected in fall before harvest, than those in spring. Significant differences for tillage treatment were not found below 30 cm depth. At high water potentials, lower porosity under no-till may restrict gaseous exchange and create conditions unfavorable for germination and seedling development. However, higher number of biochannels in the no-till soil may compensate for this reduced exchange.

Additional index words: Tillage, Soil structure, Soil physical properties.

A primary objective of research on soil tillage is to characterize the soil conditions induced by tillage operations, and determine which of the resulting conditions are most favorable for plant growth. During the past several decades, the question of how much tillage is necessary or desirable has come under increasing scrutiny by researchers is an attempt to reduce unnecessary tillage. As an outgrowth of this inqury, no-till, that is, the growing of crops in the absence of all tillage except that necessary for the placement of the seed, has developed. It has been forecast by the USDA Office of Planning and Evaluation that 35% of all cropland will be under no-till or some other form of reduced tillage by the year 2000 (unpublished report, 1974).

Data show that significant changes in soil physical properties are brought about directly and indirectly by tillage operations (Soane and Pidgeon, 1975). Bulk densities under no-till are often higher than when the soil is conventionally tilled, but some reports show no difference (Cannell and Finney, 1973). Baeumer and Bakermans (1973) have cited several researchers who found air-filled porosities of less than 10% at —100 mb water potential with no-till on medium to fine textured soils. This condition may have consequences detrimental to plants because oxygen diffusion is severely restricted at these low porosities. (Wesseling and Van Wijk, 1957). It has also been reported that increased number and continuity of pores > 1 mm diameter occur in no-till soil due to stabilized channels resulting from plant and/or animal activities, hereafter called biochannels (Ehlers, 1975). These and other structures, such as shrinkage cracks, may also be differentially affected by tillage. All such structures have the potential for affecting characteristics of soil water flow. Peterson and Dixon (1971), for example, found that a single, pencil-sized pore, representing 0.002% of the area of a 1.35 m^2 plot, increased measured infiltration rate from 6.0 to 10.5 cm/hour, an increase of 75%.

The present study was conducted to obtain data on bulk densities, water contents, air-filled porosities, hydraulic conductivities, and number of biochannels found on the contrasting treatments of conventional tillage, and no-till for continuous corn (*Zea mays* L.) on fine-textured soil. Conventional tillage consists of plowing in the fall, typically followed by one disking plus one field cultivation or, alternatively, two field cultivations.

METHODS

Description of Field Plots and Soils

This research was carried out on established field plots of a tillage experiment in continuous corn. Tillage treatments were initiated in 1969 on complete randomized plots 9 × 38 m with four replicates. The experimental area was located at the Southern Experiment Station, University of Minnesota, Waseca. The plots were on a moderately uniform, south facing slope of 2 to 4%. Tile lines ran perpendicularly to the corn rows and were spaced 23 m between adjacent lines.

The following treatments were selected for study of their physical characteristics:

1. Fall-plowing followed by spring disking and field cultivating with one post-planting cultivation.
2. No-tillage before or after planting.

Soils of the experimental area consist of LeSueur clay loam, a member of the fine-loamy, mixed mesic family of the Aquic Argiudolls with small intermixed amounts of related soils differing slightly in drainage characteristics. The mollic epipedon of these soils varies from 23 to 46 cm in depth. Particle size analysis for the plow layer determined on composite random samples obtained throughout the test site showed 36.5% sand, 11.9% coarse silt, 24.2% fine silt, and 27.4% clay.

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Soil Sampling

A truck-mounted, soil-coring sampler, 1.2-m long, fitted with a cutting tip 4.3 cm i.d. was used to collect soil samples. Samples were taken in the corn rows to eliminate the effect of traffic associated with the current year's soil preparation. Three 1-m cores from two plots per treatment, and four 1-m cores from three plots per treatment were collected in May and September 1974, respectively. Cores were discarded if excessive subsidence of the soil surface (> 2 cm) occurred during probe insertion. Cores were carefully sectioned in 10-cm lengths, and immediately sealed to prevent water loss.

Undisturbed samples were also collected using a double-cylinder sampler to remove soil cores 7.6 × 7.6-cm for measuring saturated hydraulic conductivity, air-filled porosity, numbers of visible channels, coefficient of linear extensibility (COLE), and bulk density. Six cores were taken at each of two depths (7.5 to 15 cm and 30 to 37.5 cm) from four plots per treatment in May and again in September 1974 and were stored in a refrigerator at 3 C until measurements were made.

Saturated hydraulic conductivities were determined using a constant head technique. Thereafter, samples were desorbed and equilibrated sequentially on pressure plate apparatus at pressures of —20, —50 and —100 mb, equilibration being reached at each pressure in no longer than 5 to 7 days. Porosities were determined from the weight difference between the saturated and equilibrated samples.

Coefficients of linear extensibility (Soil Conservation Staff, 1972) were determined on some samples by the difference in core lengths measured at water contents of 0.33 bar and air-dry condition divided by the air-dry length. Volume changes from —330 mb to 0 mb potential were also determined by allowing samples equilibrated at —330 mb to sorb water until no change in length occurred. After oven drying, bulk densities were calculated, based on field sample volume, and the number of visible channels on both surfaces of the sample cores were counted.

RESULTS AND DISCUSSION

Bulk densities for tillage treatments are shown in Table 1 and Fig. 1. The data indicate that under no-till, bulk densities were significantly higher within the surface 30 cm than under conventional tillage. Differences between treatments were greatest in the spring. The fact that differences were apparent only in the surface 30 cm agrees with other research showing that bulk density differences due to tillage generally converge at depths greater than 30 cm (Soane, 1970; Blake et al., 1976). These data support other findings that show a general increase in bulk densities under no-till (Cannell anl Finney 1973). By September, however, differences had reduced greatly, because of increased densities under conventional tillage.

Data on air-filled porosities at various soil water potentials are presented in Table 2. The air-filled porosities for no-tillage were never greater than for conventional tillage in spite of increased macroporosity from biochannels (Table 3). Porosities were significantly lower for no-till at the 7.5 to 15 cm depth.

Lower air-filled porosities for no-till suggest that during the spring, when soil water contents are ordinarily high, this treatment is more likely to adversely affect plant growth because of restricted gas exchange. This factor may be one reason for the claim by Triplett et al. (1973) that no-till is less successful with poorly drained soils. Other data suggest possible indirect, adverse effects of no-till on fine textured soils: greater damage to the root system by pathogenic fungi (Van Doren et al., 1976) and by toxic gases (Smith and Jackson, 1974).

Air-filled porosity would also be expected to vary

Table 1. Bulk densities of undisturbed cores (7.6 × 7.6 cm); 1974.

Depth	No-till	Fall plow	SE (6 df)†	p‡
cm	g/cm³			
		20 May		
7.5–15	1.32	1.05	0.02	<0.001
30 –37.5	1.29	1.29	0.02	ND
		20 September		
7.5–15	1.25	1.12	0.03	0.02
30 –37.5	1.29	1.27	0.02	ND

† Standard errors of tillage means (SE = S/24).
‡ Probability the null hypothesis, H_o:$\mu_{difference}$ = o. ND means no difference (P > 0.20). Mean values for a given tillage-depth-season are derived from 24 samples.

Table 2. Air-filled porosities of undisturbed cores (7.6 × 7.6 cm); 1974.

Soil water potential	20 May No-till	20 May Fall plow	20 May SE (6 df)	20 May P	20 September No-till	20 September Fall plow	20 September SE (6 df)	20 September P
mb	cm³/cm³							
			7.5–15 cm depth					
–20	0.051	0.118	0.012	0.01	0.094	0.122	0.012	0.16
–50	0.072	0.176	0.015	0.01	0.123	0.172	0.016	0.08
–100	–	–	–	–	0.143	0.198	0.016	0.05
O.D.†	0.47	0.57	0.013	0.01	0.502	0.554	0.013	0.03
			30–37.5 cm depth					
–20	0.061	0.051	0.011	ND	0.056	0.060	0.007	ND
–50	0.096	0.086	0.013	ND	0.088	0.100	0.009	ND
–100	–	–	–	–	0.109	0.121	0.009	ND
O.D.†	0.47	0.47	0.012	ND	0.481	0.487	0.009	ND

† Porosity determined on samples oven-dried at 105 C.

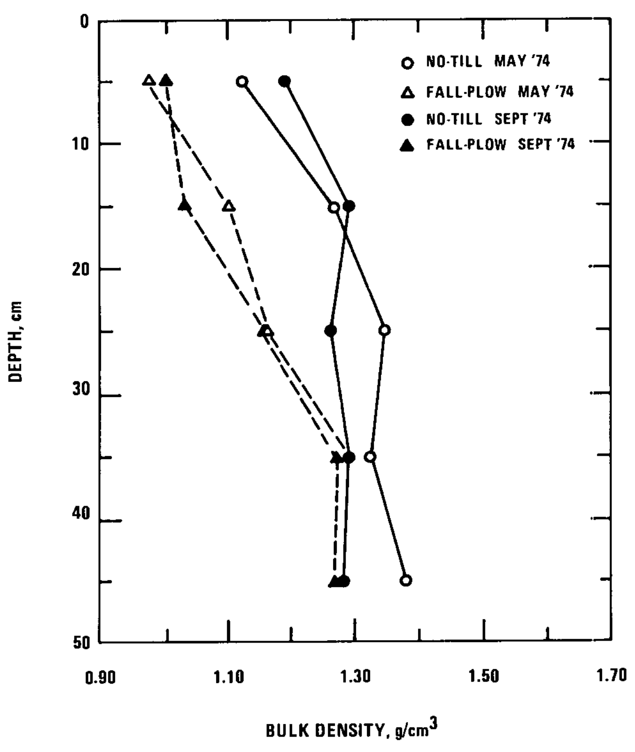

Fig. 1. Bulk density profiles for no-till and fall-plow plots, LeSueur clay loam.

Table 3. Geometric mean numbers of biochannels of different sizes for no-till and conventional tillage.

Channel diam.	No-till	Fall plow	No-till	Fall plow	SE (6 df)	P	No-till	Fall plow	No-till	Fall plow	SE (6 df)	P
mm	Channels/m²		ln (Channels/m²)†				Channels/m²		ln (Channels/m²)			
					7.5–15 cm depth							
>6	112	58	4.934	4.445	0.260	0.18	69	14	4.565	3.710	0.303	0.16
2-3	366	136	5.973	5.093	0.346	0.09	105	45	4.889	4.274	0.259	0.15
1-2	799	234	6.693	5.568	0.920	0.05	217	100	5.502	4.844	0.432	ND
Total	1,317	420	7.203	6.103	0.345	0.07	635	216	6.496	5.492	0.151	0.01
					30–37.5 cm depth							
>3	133	87	5.073	4.739	0.166	0.13	128	27	5.049	3.991	0.429	0.14
2-3	492	390	6.253	6.033	0.298	ND	252	166	5.634	5.263	0.236	ND
1-2	968	846	6.903	6.773	0.176	ND	660	481	6.533	6.233	0.203	ND
Total	1,699	1,443	7.453	7.293	0.168	ND	1,247	767	7.150	6.677	0.205	0.18

† For conversion use ln (Channels/m² + 27) = Channels/m². The area conversion factor 27 was used since actual counts were made on surfaces of undisturbed cores 7.6 × 7.6 cm.

with volume change. Owing to relatively high clay content, samples exhibited volume change with change in water content. Consequently two estimates of volume change were made. Coefficients of linear extensibility were calculated on samples taken in the spring to determine if differences between treatments existed. The mean values were 0.046 and 0.042 respectively for the 7.5 to 15-cm and 30 to 37.5-cm depths under no-till; 0.088 and 0.041 for the equivalent depths under conventional tillage. A comparison of mean COLE values shows that those at the 7.5 to 15 cm depth for conventional tillage were significantly larger in a statistical sense than at other tillage depths. In order to assess possible bias from calculating air-filled porosity on the basis of original sample volume, change in linear dimension of samples from both treatment during sorption of water from −330 to zero mb water potential was also measured. Data show a 5% volume increase for fall-plow samples and a 2% volume increase for no-tillage samples. Assuming normal volume change, the maximum bias between treatments would be around 1% porosity in the values in Table 2 at −100 mb potentials.

Numbers of biochannels are given in Table 3. These consisted mostly of earthworm channels, but we made no attempt to distinguish between channels caused by earthworms or decomposing roots. Shrinkage cracks and other planar voids were not included in these counts.

Counts in no-till samples were two to four times greater than in conventionally tilled samples. There were 600 to 1,700 visible channels/m². If 1,000/m² were uniformly distributed the maximum distance between channels would be only 3.2 cm. The effect on porosity due to these channels would be small, about 0.25%, if they were assumed to be vertically homogeneous. The effect of such a network on root growth and fluxes of water, gases, and heat would be expected to be significant.

The counts reported in Table 3 tend to be higher than others have found (Barley, 1961; Teotia et al., 1950; Ehlers 1975; Russell, 1973). This result may be partly due to the fact that we included smaller-sized channels (< 2 mm), which included a number of root channels in our counts. Also, several of estimates cited by Barley (1961) and Russell (1973) included only counts of channels at the soil surface where number and diameter estimation are difficult, because of unstable soil. The fact that the 7.5 to 15 cm samples of no-till soil consistently showed higher counts than the fall-plowed soil samples is most likely related to the earthworm activity encouraged by the greater amount of organic mulch. Also, one would anticipate fewer channels under the plowed treatment because of destruction after shearing the furrow slice. Comparison of mean counts between fall and spring suggests that this was not so. A high activity of channel formation in spring seems to compensate for this destruction of continuous pores.

Relatively little has been done to characterize the effect continuous channels would have on soil aeration. Satchell (1967) opined that the effect of earthworms on soil aeration is of only minor importance. However, recent work by Dowdell and Crees (1976) showed that oxygen contents at 15 cm depths under no-till are typically greater than under plowing during the growing season. They tentatively attributed this result to the system of uninterrupted pores created by earthworms under no-till. Though our data do not resolve the issue, they show an overall reduced porosity in no-till, fine-textured soil. Nevertheless, biochannels of the frequency shown in our data may have a greater influence on aeration than suggested by their actual volume, because the pores are less tortuous than the usual random assemblage found between soil particles and aggregates. Also, the pore geometry is a simpler one.

Statistically we found a trend for the coefficient of variation, that is the ratio of the standard deviation to the mean, to remain constant over treatments of different mean value. This finding suggests that treatment effects are likely to be multiplicative rather than additive (Snedecor and Cochran, 1972). For this reason, the data in Table 3 were transformed to logarithm of number of channels +0.25 for presentation and analysis. Geometric means are also presented. A numerical value of 0.25 was added to each count to remove zero values from the data. In general, the transformation tended only slightly to reduce the probability levels for treatment effects, in contrast to nontransformed data.

Water contents under the tillage treatments are presented in Figs. 2 and 3. Gravimetric water contents were not significantly different at either sampling date. The apparent tillage × depth interaction of water contents observed in the 5 to 15 cm depth during spring is primarily due to low water contents in the fall plow treatment (Fig. 2). This result can be attributed to the absence of surface residues and the loose condition of the soil on the fall-plowed treatment, which promoted rapid drying in the surface 10 cm.

No-tillage resulted in a significantly higher volumetric soil water content than conventional tillage in the surface 30 cm (Fig. 3). At greater depths the treatments were not significantly different. The significance of the increased water content as a factor in yields under no-till may be of little importance if, as van Ouwerkerk and Boone (1970) have shown, additional water is not available for plant growth because of a finer pore system under no-till.

Hydraulic conductivities for tillage treatments are shown in Table 4. There was a significant positive correlation between conductivities and bulk density and a significant negative correlation with porosities at high water potential. In the 30 to 37.5 cm depth the significance of these correlations decreased owing to greater soil homogeneity within treatments, and a significant correlation developed between conductivity and number of earthworm burrows.

It will be noted that conductivities were greater in September than in May for both no-till and conventional tillage at the 30 to 37.5 cm depth. This finding is surprising if one assumes that the greater number of earthworm burrows in the spring samples (Table 3) could conduct larger quantities of free water. It is suggested that soils were relatively less hydrated in September than in May. Even though samples were wetted according to standard procedure (Klute, 1965), Emerson (1955) has shown that clays may require weeks, or even months, to rehydrate once dried to —15 bars. Conductivities determined on fall samples may, therefore, be greater because of persistence of small shrinkage cracks or fissures in the soil matrix present from summer dehydration.

SUMMARY AND CONCLUSIONS

This experiment provides evidence that clay loam soils managed under no-till acquired physical characteristics quite different from those found under conventional tillage. Generally, differences in characteristics between treatments were limited to surface samples. Differences were either smaller or undetectable for samples collected in fall prior to harvest than for those taken in the spring.

Data show that soil under no-till had significantly greater bulk densities both spring and fall in the plow zone. Values ranged from 1.24 to 1.32 g/cm^3 for no-till compared to 1.05 to 1.12 g/cm^3 for conventional tillage. This increase occurred in spite of the fact that mean number of channels 1 mm in diameter and larg-

Table 4. Saturated hydraulic conductivities of undisturbed cores (7.6 × 7.6 cm); 1974.

Depth	No-till	Fall-plow	No-till	Fall-plow	SE (6 df)	P
cm	— cm/hour —		——ln (cm/hour)——			
			20 May			
7.5–15	14.6	38.2	2.68	3.64	0.201	0.02
30 –37.5	19.9	19.4	2.99	2.76	0.121	ND
			20 September			
7.5–15	65.0	40.4	4.18	3.70	0.334	ND
30 –37.5	43.4	50.5	3.78	3.92	0.225	ND

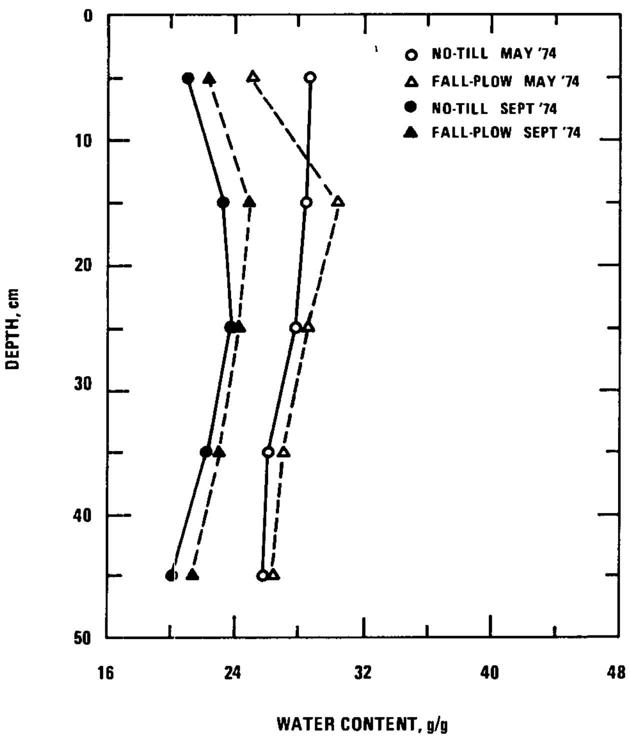

Fig. 2. Water content profile for no-till and fall-plow plots, LeSueur clay loam.

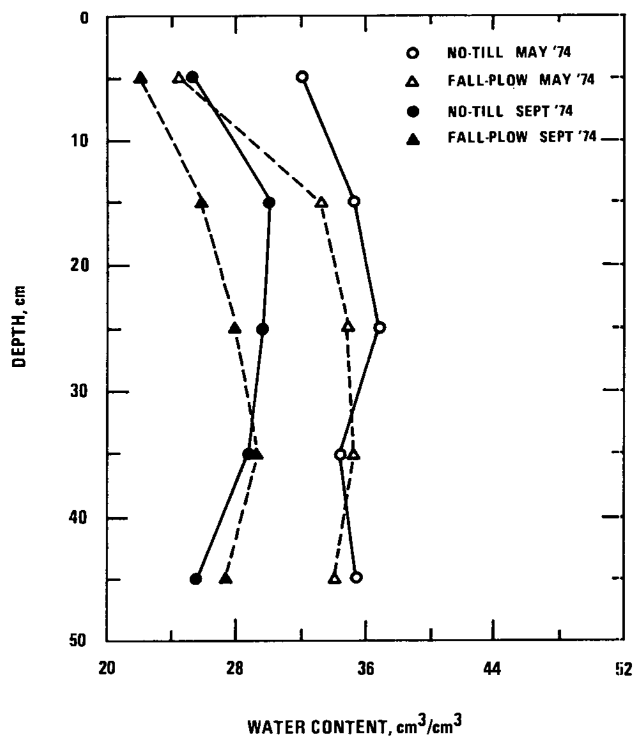

Fig. 3. Volumetric water content for no-till and fall-pot plots, LeSueur clay loam.

er, created by earthworms and decomposing rootlets, were significantly greater for no-till. Counts for no-till ranged from 666 to 1,732/m^2 compared to 243 to 1,475/m^2 for conventional tillage.

A consequence of the increased density was that air-filled porosites under no-till were less than for conventional at all potentials measured. At -100 mb, no-tillage averaged 0.143 cm^3/cm^3 compared to 0.198 cm^3/cm^3 for conventional tillage. In addition, saturated hydraulic conductivities of surface samples collected in May were lower under no-till than conventional averaging 14.6 cm/hour and 38.2 cm/hour, respectively.

Volumetric water contents were greater under no-till than under conventional tillage, values ranging from 0.28 to 0.35 cm/cm and from 0.25 to 0.31 cm/cm, respectively. If this is attributed simply to an increase in water retaining pores, then, as van Ouwerkerk and Boone (1970) have shown, it may be of little importance for plants. However, an increased infiltration and lowered evaporation rate owing to a surface mulch peculiar to no-till has been shown (Cannell and Finney, 1973). Furthermore, Ehlers and Baeumer (1974) and Ehlers (1975) found that biochannels conduct water past surface layers where it is protected from evaporation.

At times when there is excessive water, that is, high soil water potentials, reduced porosity under no-till may restrict gaseous exchange and create suboptimal conditions for germination and seedling development. However, biochannels of the frequency found in our data may compensate for this reduced exchange since the pores are relatively less tortuous and simpler in pore geometry than in soils with fewer channels. Further research is needed to resolve these issues.

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